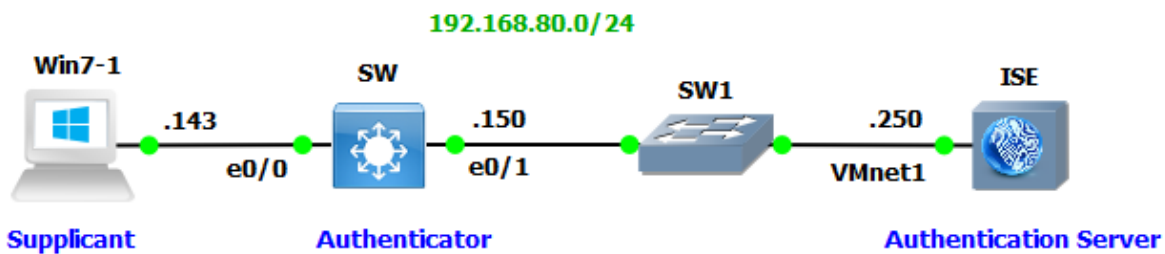


Profile Lab Topology:



Dot1x Configuration

```
SW(config)#interface vlan 1
SW(config-if)#ip add 192.168.80.150 255.255.255.0
SW(config-if)#no shutdown
SW(config-if)#exit
SW(config)# aaa new-model
SW(config)# username admin password 123
SW(config)# radius server myserver
SW(config-radius-server)#address ipv4 192.168.80.250
SW(config-radius-server)#key test123
SW(config)# aaa authentication dot1x default group radius
SW(config)#aaa authorization network default group radius
SW(config)#aaa accounting dot1x default start-stop group radius
SW(config)#radius-server vsa send
SW(config)#radius-server vsa send accounting
SW(config)#radius-server vsa send authentication
SW(config)#radius-server attribute 6 on-for-login-auth
SW(config)#radius-server attribute 8 include-in-access-req
SW(config)#radius-server attribute 25 access-request include
SW(config)#ip device tracking
SW(config)# dot1x system-auth-control
SW(config)# interface e0/0
SW(config-if)# switchport host
SW(config-if)# switchport access vlan 1
SW(config-if)# authentication port-control auto
SW(config-if)# dot1x pae authenticator
SW(config-if)# authentication host-mode multi-host
SW(config-if)#authentication timer reauthenticate server
SW(config-if)#authentication event fail action next-method
SW(config-if)#authentication periodic
SW(config-if)#authentication violation shutdown
SW# show dot1x interface e0/0
SW# show dot1x all
```

RADIUS Probe:

- o RADIUS Probe is one of the most common probes using by ISE.
- o ISE profile based on RADIUS attributes collected from RADIUS messages.
- o RADIUS probe also listens to CDP & DHCP attributes send in RADIUS accounting packets.
- o Such as User-Name, Calling-Station-Id, NAS-IP-Address, NAS-Port & Framed-IP-Address.
- o There is a lot of more information that can be pulled from the RADIUS attributes.

Add or Remove Columns ▾ Refresh Reset Repeat Counts					
Status	Details	Repeat Count	Identity	Endpoint ID	Endpoint Profile
All ▾					
i2		2	admin1	F0:6E:0B:12:34:56	Microsoft-Workstation
i0			admin1	F0:6E:0B:12:34:56	Unknown

Refresh + Add Trash ▾ Edit MDM Actions ▾ Refresh MDM Partner Endpoint Import ▾ Export ▾							
Endpoint Profile	MAC Address	Vendor(OUI)	Logical Profiles	Hostname	MDM Server	Device Identifier	IP Address
✕ Endpoint Prof	MAC Address			Hostname	MDM Serv	Device Id	IP Address
☐ Microsoft-Workstation	F0:6E:0B:12:34:56	UNKNOWN		WIN7			192.168.80.1...

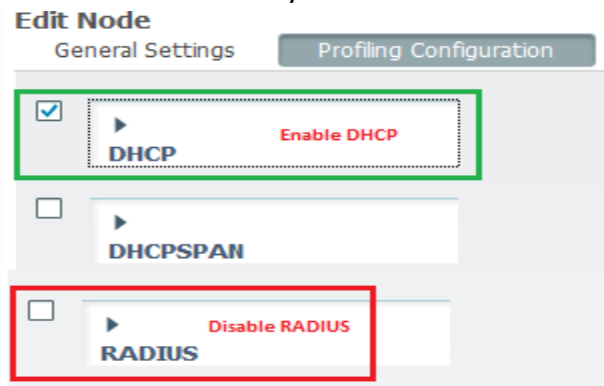
Looking at one of profiled hosts **Administration>Identity Management>Identities>Endpoints**, see the attributes that were collected:

Called-Station-ID	AA-BB-CC-00-01-00
Calling-Station-ID	F0-6E-0B-12-34-56
DestinationIPAddress	192.168.80.250
DestinationPort	1645
Device IP Address	192.168.80.150
Device Type	Device Type#All Device Types
DeviceRegistrationStatus	NotRegistered
ElapsedDays	0
EndPointMACAddress	F0-6E-0B-12-34-56
EndPointPolicy	Microsoft-Workstation
EndPointProfilerServer	ISE1.google.com
EndPointSource	RADIUS Probe
Framed-IP-Address	192.168.80.148
IdentityGroup	Workstation

DHCP Probe:

- o DHCP Probe collect DHCP request attribute from user, proxy & Helper.
- o Traffic forwarded to Cisco ISE with helper address or SPAN.
- o DHCP Probe using ip helper-address interface configuration command.
- o Cisco ISE will only use the incoming DHCP data to profile endpoints.

Navigate to **Administration > System > Deployment > Profiling Configuration** this time disable RADIUS Probe and only enable DHCP Probe.



Configure the switch to forward DHCP request to Cisco ISE.

```
SW(config)#interface vlan 1
SW(config-if)#ip helper-address 192.168.80.1
SW(config-if)#ip helper-address 192.168.80.250
```

Let us look again the profiled hosts **Administration > Identity Management > Identities > Endpoints**, see the attributes that were collected this time by DHCP.

EndPointPolicy	Microsoft-Workstation
EndPointProfilerServer	ISE2.google.com
EndPointSource	DHCP Probe
IdentityGroup	Workstation
MACAddress	00:0C:29:55:68:E0
MatchedPolicy	Microsoft-Workstation
OUI	VMware, Inc.
PolicyVersion	0
StaticAssignment	false
StaticGroupAssignment	false
Total Certainty Factor	30
UpdateTime	1562138211055
chaddr	00:0c:29:55:68:e0
ciaddr	0.0.0.0
client-fqdn	WIN7

SNMPQUERY Probe:

- o SNMP Query Probe send query packets known as SNMP GET Requests.
- o System Queries are periodic depending upon polling interval.
- o System Queries collect Bridge, IP CDP Cache Entry LLDP Local System Data.
- o Interface Queries are generated when RADIUS Accounting Starts.
- o Interface Queries are generated when SNMP detect linkup.

☒ **SNMPQUERY**

Retries: 2

Timeout: 1000

EventTimeout: 30

Description: This probe collects details from network devices such as Interface, CDP, LLDP and ARP.

SNMP Trap Probe:

- o SNMP Trap receives information from configured NAD support MAC notification.
- o SNMP Trap receives info form configured NAD support linkup, linkdown & informs.
- o For SNMPTRAP to be functional, you must also enable the SNMPQUERY probe.
- o To make this feature functional, configure the NAD to send SNMP traps or informs.
- o SNMP Trap probe receives information from the specific network access devices.
- o When ports up or go down & endpoints disconnect from or connect to network.

☒ **SNMPTRAP**

Link Trap Query: ☒

MAC Trap Query: ☒

Interface: GigabitEthernet 0

Port: 162

Description: This probe receives Linkup, Linkdown and MAC notification traps from network devices.

Go to **Administration > Network Resources > Network devices** click on network device in my case switch go down to **SNMP Settings** enable **SNMP Settings**, Choose version **2c** and type SNMP Community **123**. Click **Save** button to save the settings.

SNMP Settings

* SNMP Version: 2c

* SNMP RO Community: 123

SNMP Username:

Security Level:

Auth Protocol:

Auth Password:

Privacy Protocol:

Privacy Password:

* Polling Interval: 600 seconds (Valid Range 600 to 86400 or zero)

Link Trap Query: ☒

MAC Trap Query: ☒

* Originating Policy Services Node: Auto

Configuration of SNMP on Switch

S1(configure)# snmp-server community **123** rw

S1(configure)# snmp-server enable traps

S1(configure)# snmp-server host 192.168.8.250 **123**

Let us look again the profiled hosts **Administration>Identity Management> Identities> Endpoints**, see the attributes that were collected this time by **SNMPQuery**.

Attribute List

BYODRegistration	Unknown
DeviceRegistrationStatus	NotRegistered
ElapsedDays	0
EndPointPolicy	Cisco-Device
EndPointProfilerServer	ISE1.google.com
EndPointSource	SNMPQuery Probe
IdentityGroup	Profiled
InactiveDays	0
MACAddress	AA:BB:CC:80:01:00